

01_Project Charter

1. Project Title

Olist E-Commerce Case Study: End-to-End Business Intelligence & Analytics Project

2. Industry

E-Commerce / Retail Analytics

3. Business Problem

The e-commerce company is experiencing inconsistent revenue growth, declining repeat purchases, and operational inefficiencies in areas such as deliveries and customer satisfaction. Leadership wants to identify the key customer, product, and operational factors affecting long-term revenue growth and customer retention. The company aims to use data-driven insights to improve customer experience, optimize business performance, and increase repeat customer value.

4. Project Objective

The objective of this project is to analyze customer behavior, product performance, payment patterns, delivery efficiency, and customer satisfaction using the Olist e-commerce dataset in order to uncover actionable insights that can help improve customer retention, optimize operations, and drive sustainable revenue growth.

Why This Problem Matters

In e-commerce businesses, acquiring new customers is expensive. Long-term profitability depends heavily on retaining customers, improving operational efficiency, and understanding purchasing behavior. Poor delivery experience, low customer satisfaction, or underperforming products can directly impact repeat purchases and overall revenue growth.

5. Project Workflow

The project follows an end-to-end analytics workflow starting from raw dataset inspection and preprocessing to business analysis and dashboard reporting.

Workflow:

Raw Dataset → Python Data Cleaning & Validation → PostgreSQL Database Integration → SQL Business Analysis → Power BI Dashboarding → Business Recommendations

6. Key Goals

- Analyze customer purchasing behavior and repeat purchase patterns
- Identify high-value customers and revenue-driving segments
- Evaluate product and category performance
- Study customer satisfaction and review trends
- Analyze delivery performance and operational inefficiencies
- Identify factors contributing to customer churn and low retention
- Generate business recommendations using data-driven insights
- Perform data cleaning and validation using Python
- Build an interactive Power BI dashboard for executive reporting

7. Expected Deliverables

- Cleaned and validated analytical dataset
- PostgreSQL relational database
- SQL-based business analysis
- Python preprocessing and EDA notebooks
- Power BI executive dashboard
- Business recommendations
- Final analytical report
- GitHub portfolio project

8. Tools & Technologies

Purpose	Tool
Initial Inspection	Excel
Data Cleaning & Processing	Python (Pandas, NumPy, Jupyter Notebook)
Database Management	PostgreSQL
SQL Environment	pgAdmin
Business Analysis	SQL
Visualization	Power BI
Documentation	Notion
Version Control	GitHub
Development Environment	VS Code

8. Dataset Source

Olist Brazilian E-Commerce Dataset

<https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>

9. Analytical Dataset Overview

The Olist Brazilian E-Commerce dataset contains transactional and operational data from a real-world online marketplace. The dataset includes information related to customers, orders, products, sellers, payments, reviews, and geolocation details. The data follows a relational structure that enables customer behavior analysis, revenue analysis, operational performance tracking, and customer satisfaction evaluation. The dataset is suitable for end-to-end analytics, business intelligence, SQL analysis, and dashboard reporting.

10. Core Analytical Areas

- Customer Retention & Repeat Purchase Behavior
- Revenue & Payment Analysis
- Product & Category Performance
- Customer Satisfaction & Review Trends
- Delivery & Operational Performance
- Geographic & Regional Analysis

- Seller Performance Analysis
- Customer Segmentation & High-Value Customers

Final Result

The final outcome of this project is an end-to-end business intelligence and analytics solution for the Olist e-commerce business. The project combines Python-based data cleaning and preprocessing, PostgreSQL database integration, SQL-driven business analysis, and Power BI dashboard development to uncover actionable business insights. The analysis aims to identify operational inefficiencies, customer behavior patterns, revenue-driving factors, and customer satisfaction trends in order to support data-driven business decision-making and long-term growth strategies.